

PROJECT

Classifying Human Activities in Videos
UCF101 Dataset



TEAM CaipirinhAI

**Bastian ARDILLON
Octave SAVEAUX
Minh-Duc PHAN**

1. Introduction



Objective : Automatically classify human activities in videos using only extracted images.

12 / 101 Class

BabyCrawling

HulaHoop

Basketball

JumpingJack

Billiards

Nunchicks

BlowingCandles

PizzaTossing

CuttingInKitchen

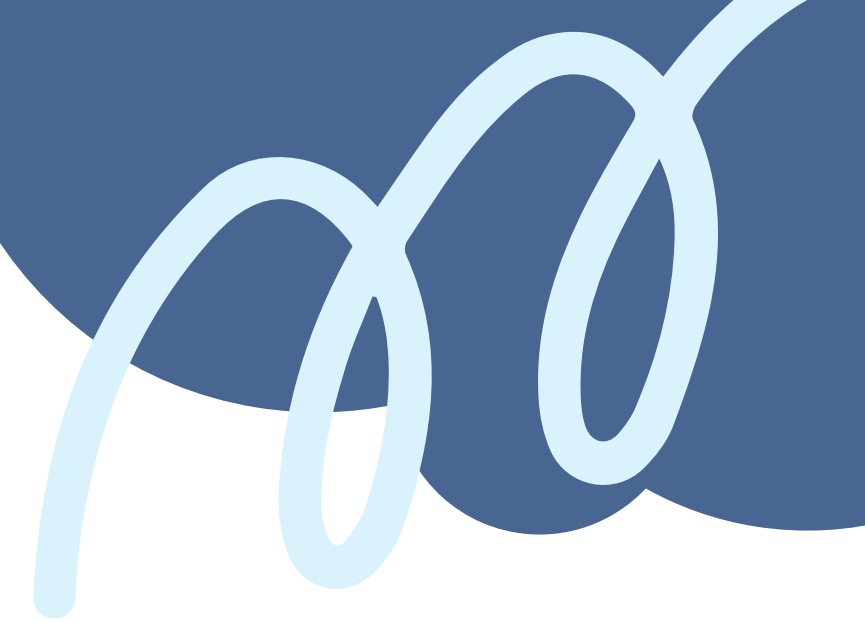
Skiing

Fencing

SumoWrestling



2. Preprocessing



Extraction

**14 frames per
video**

Resize

**1 frame →
224x224 px**

Normalization

Pixel / 255.0

Split data

**70% train ;
15% validation ;
15% test**



3. Architecture & Transfer Learning *

ResNet50

Residual – good baseline

24.1M parameters

EfficientNetB0

Best trade-off

4.38M parameters

ConvNext

Modern architecture, high precision

87.8M parameters

Video Classification Strategy.

- Extraction of 14 frames per video
- Frame-level prediction via CNN
- Probability averaging → final class (majority voting / average softmax)



4. Result

ResNet50

99.79%

EfficientNetB0

99.69%

ConvNeXt

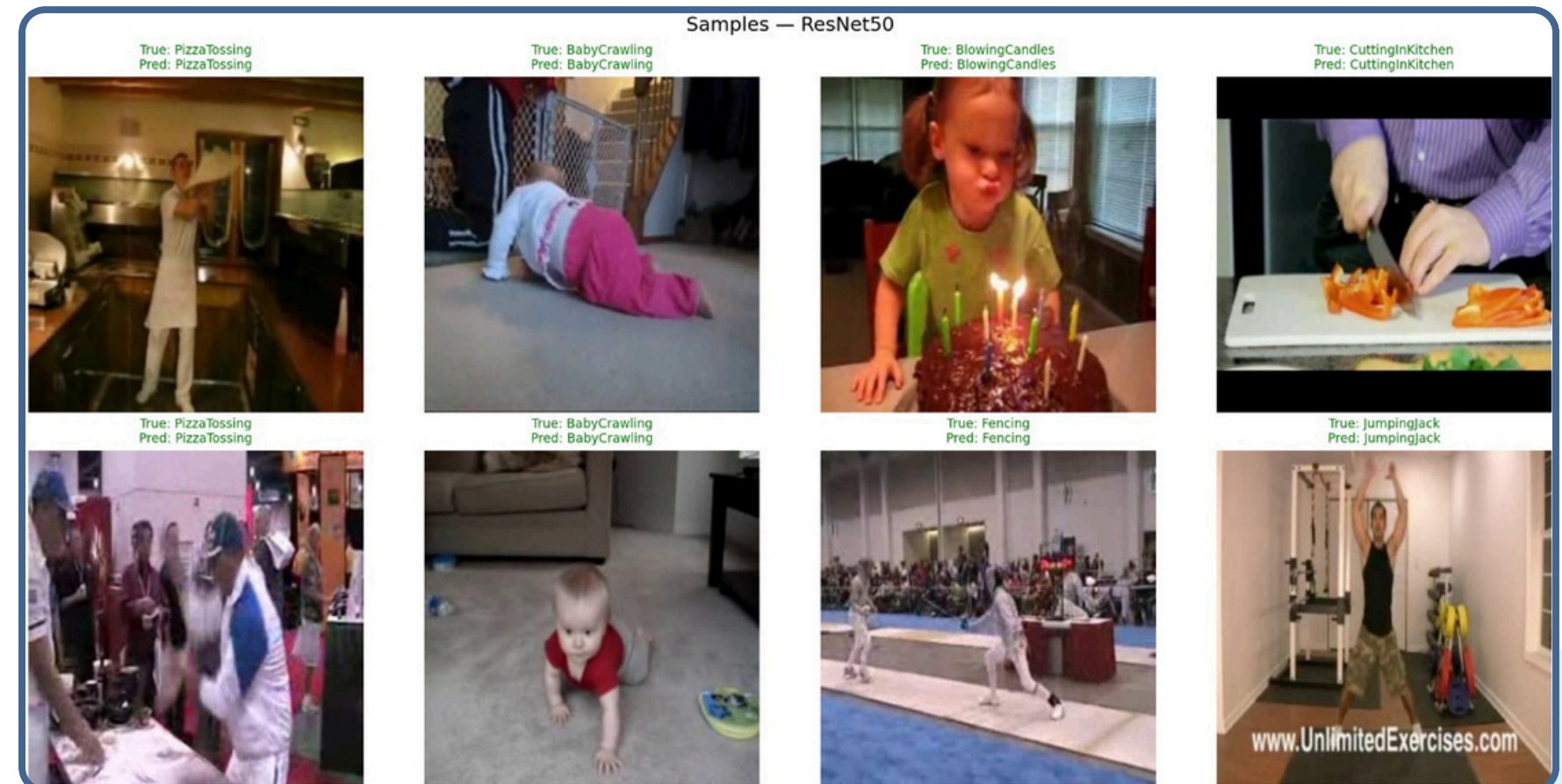
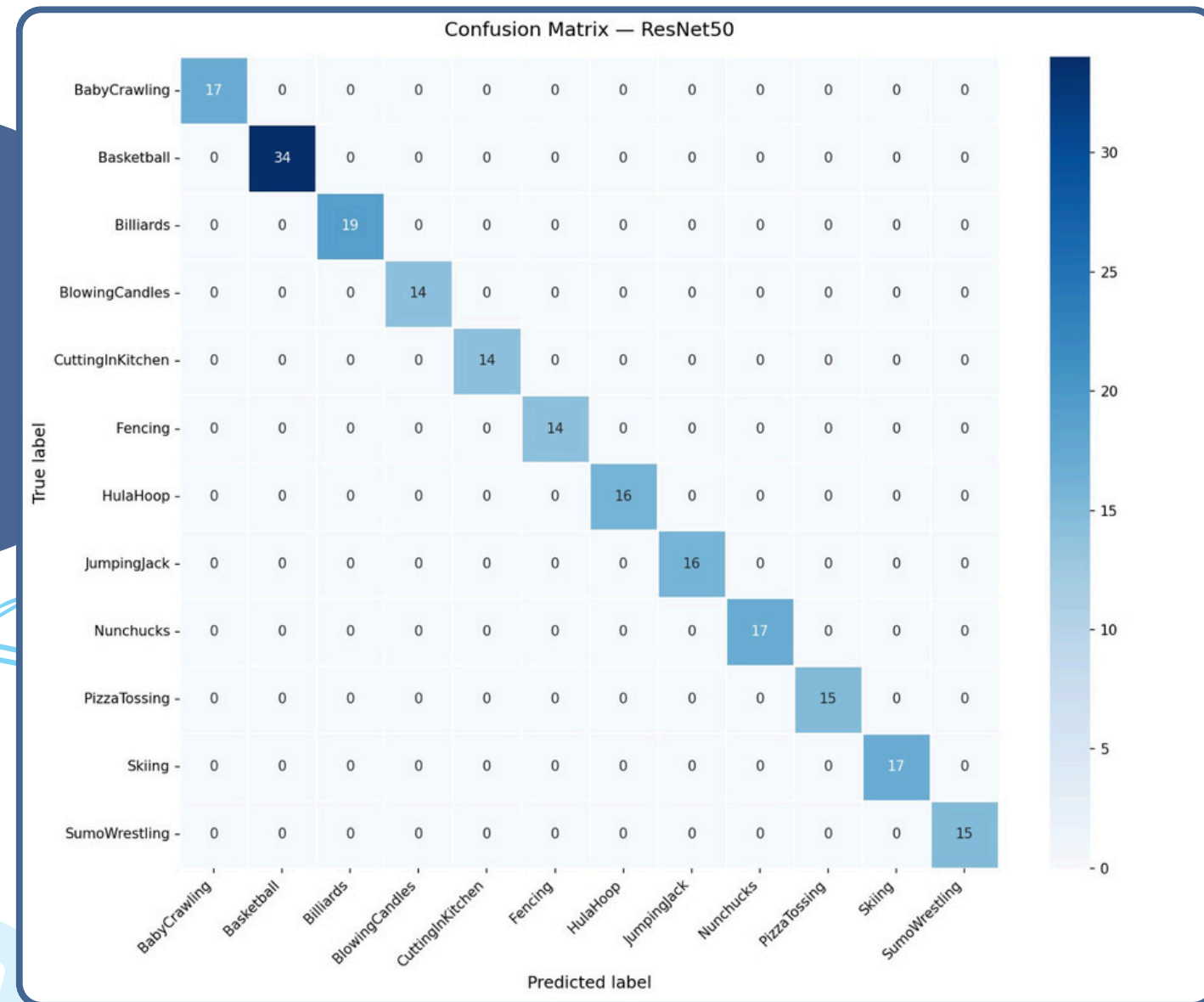
81.7%





4. Result

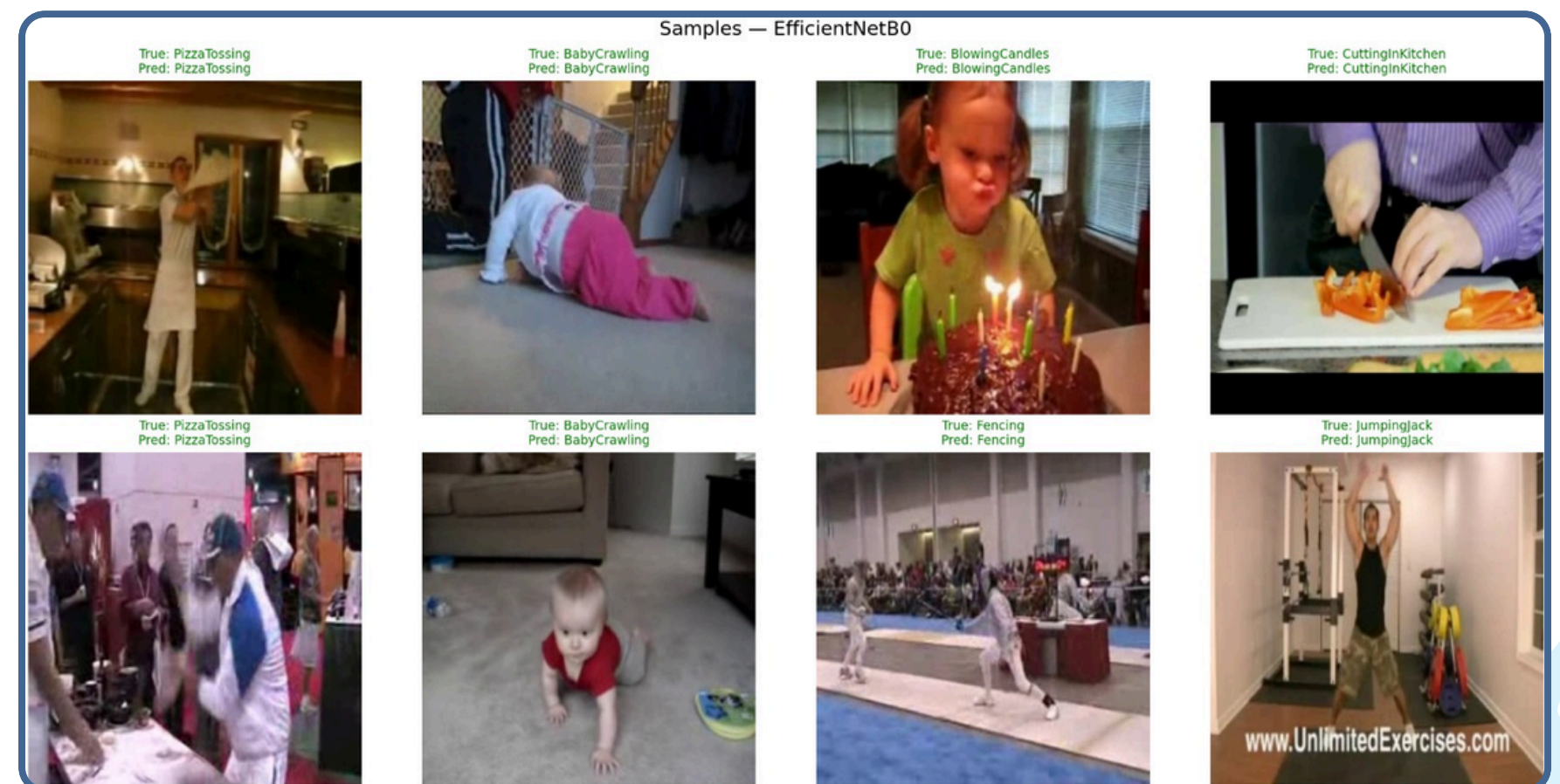
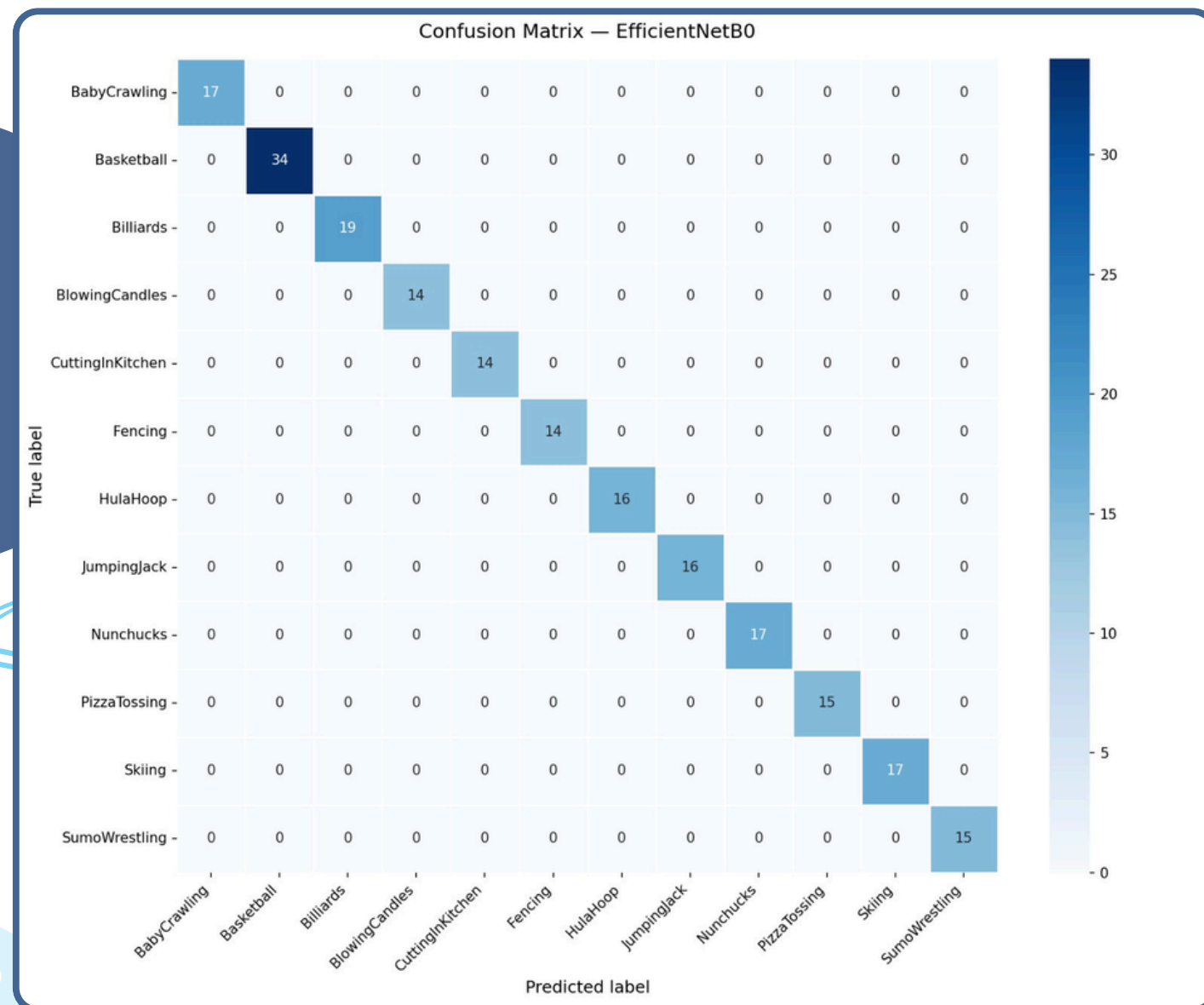
ResNet50





4. Result

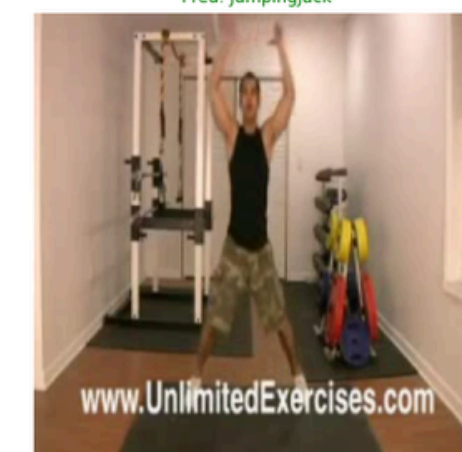
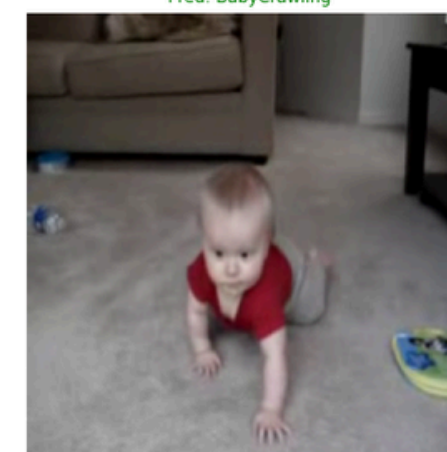
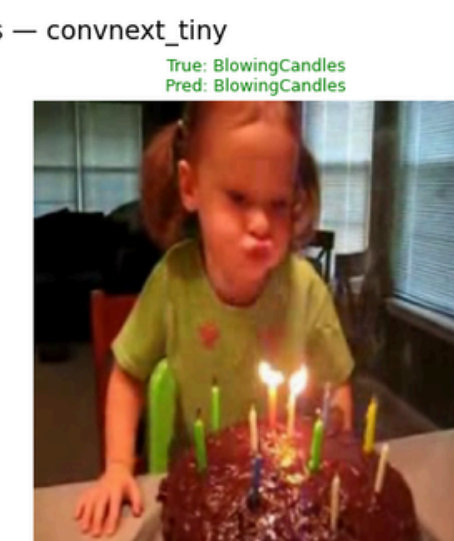
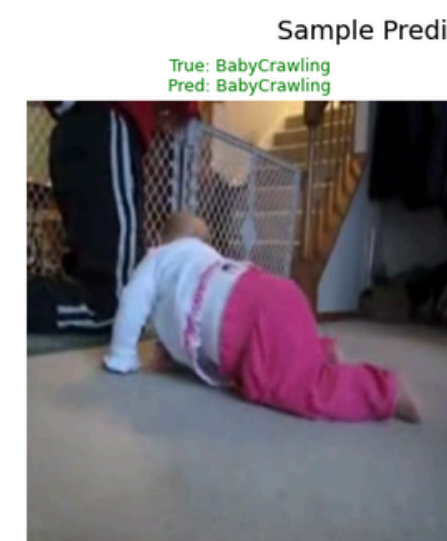
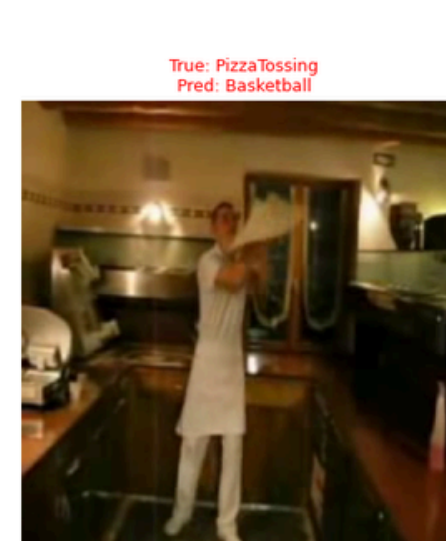
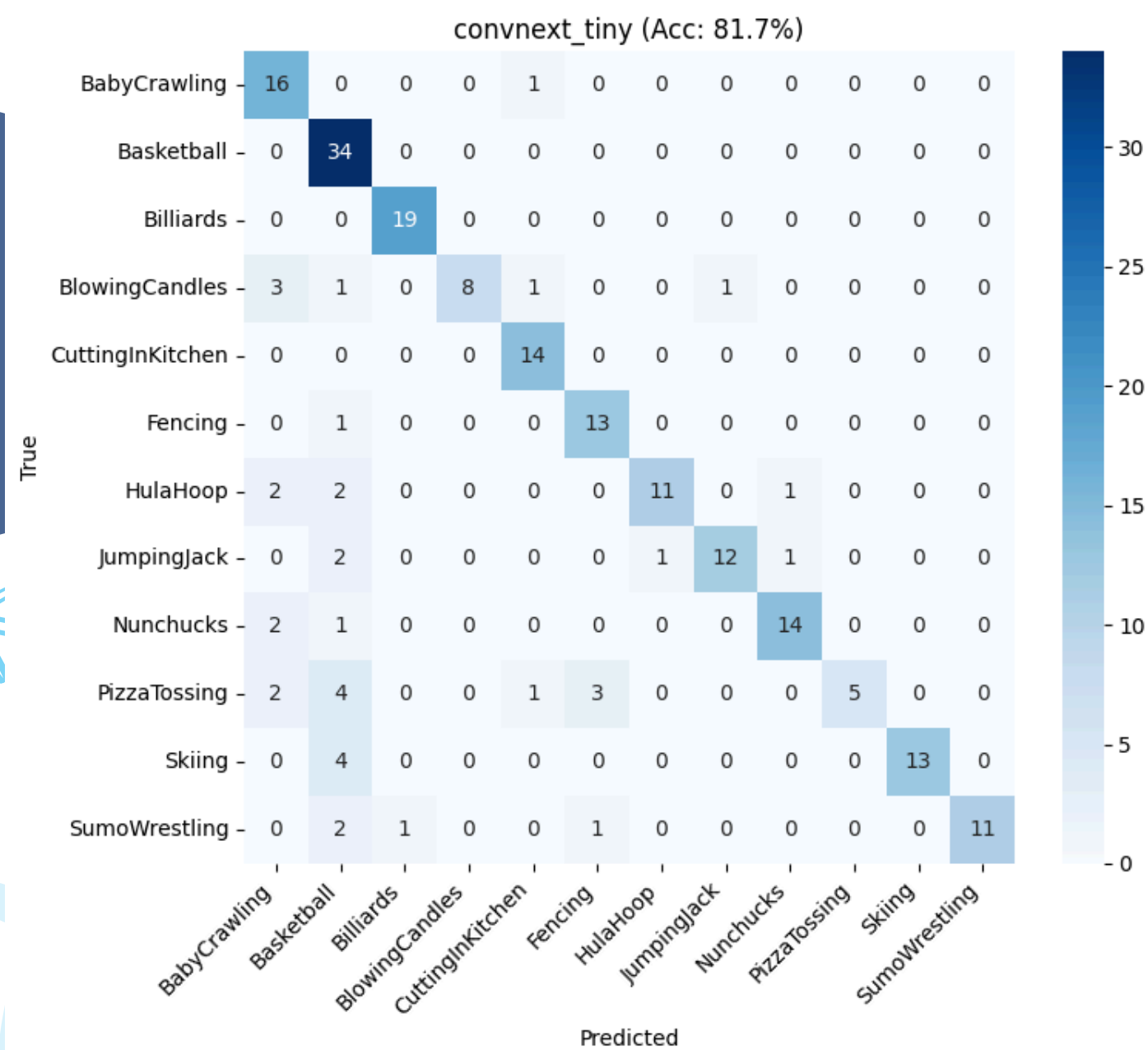
EfficientNetB0

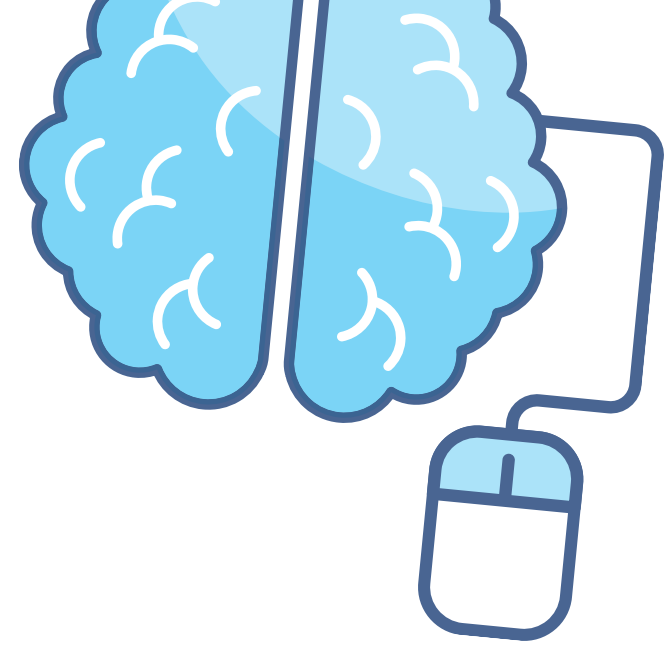




4. Result

ConvNeXt





5. Lessons Learn



What work :

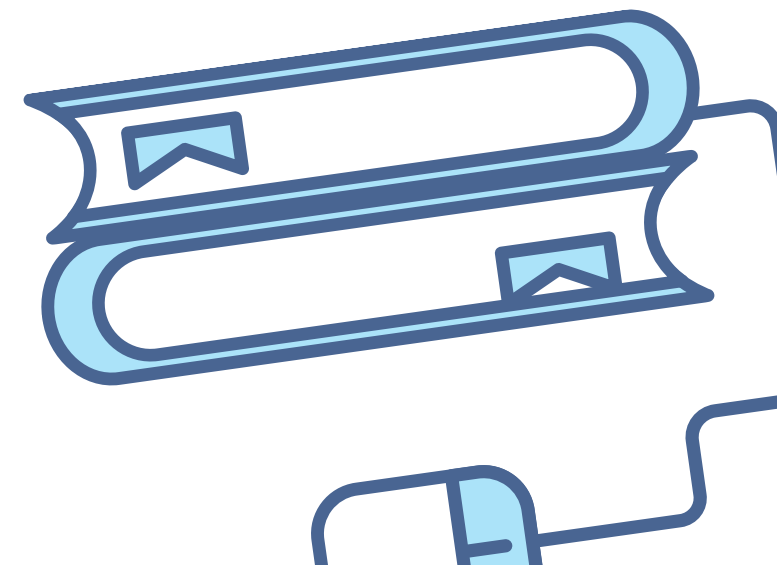
- ✧ Our two phase training pipeline are robust
- Frame averaging stabilizes predictions
- Using pre-trained ImageNet models gave fast convergence and good feature reuse on small dataset

Areas for improvement :

- ✧ Confusion between certain classes (e.g., Pizza tossing vs. Basket Ball)
- Long training times without a dedicated GPU or with bad network using kaggle
- Improve the number of categories
- Updating our pipeline with complet ConvNeXt model

For the futur project : Temporal modeling (RNN)

• Optical Flow • Data augmentation





Thank you for
your attention

